# CAPSTONE PROJECT REPORT

## *Employee Attrition Prediction System*

A Machine Learning Classification Model to Predict Employee Attrition
and Identify Key Drivers of Employee Turnover

**Dataset:** HR-Employee-Attrition.csv | 1,470 rows | 35 columns
**Models Trained:** Naive Bayes | Decision Tree | Random Forest
**Best Model:** **Random Forest — Accuracy: 85%**

2026

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# Employee Attrition Prediction System

*Capstone Project Report — 2026*

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# 1. Data Understanding Report

## 1.1 Problem Statement

We are data scientists at a company and our task is to build a classification model to predict employee attrition and identify the key drivers influencing employee turnover.

**The three core objectives of this project are:**

- Predict Attrition (Yes / No) for each employee
- Explain WHY employees leave — surface the top contributing factors
- Suggest business solutions and retention strategies based on findings

## 1.2 Business Context

Employee attrition is a major operational and financial burden for any organisation. High turnover means increased recruitment costs, loss of institutional knowledge, lower team morale, and reduced productivity. A data-driven predictive model allows HR teams to shift from reactive to proactive — catching at-risk employees before they resign.

## 1.3 Target Variable

The target variable in the dataset is Attrition:

- 1 → Yes (Employee left the company)
- 0 → No (Employee is still with the company)

## 1.4 Dataset Overview

The dataset used is HR-Employee-Attrition.csv — a standard HR analytics dataset containing detailed employee profiles.

| Property | Value |
|---|---|
| File name | HR-Employee-Attrition.csv |
| Total rows | 1,470 employees |
| Total columns | 35 features |
| Target column | Attrition (Yes / No) |
| Null values | None — dataset is complete with no missing values |
| After outlier removal | Reduced rows (IQR method applied to all numerical columns) |

## 1.5 Input Variables

The 35 columns (before dropping 4 non-informative ones) include a mix of numerical and categorical features:

| Feature | Type | Description |
|---|---|---|
| Age | Numerical | Employee age |
| DailyRate | Numerical | Daily rate of pay |
| DistanceFromHome | Numerical | Distance from home to workplace (km) |
| Education | Numerical (1–5) | Education level (1=Below College … 5=Doctor) |
| EnvironmentSatisfaction | Numerical (1–4) | Satisfaction with work environment |
| HourlyRate | Numerical | Hourly rate of pay |
| JobInvolvement | Numerical (1–4) | Degree of job involvement |
| JobLevel | Numerical (1–5) | Seniority level |
| JobSatisfaction | Numerical (1–4) | Job satisfaction rating |
| MonthlyIncome | Numerical | Monthly salary (USD) |
| MonthlyRate | Numerical | Monthly rate of pay |
| NumCompaniesWorked | Numerical | Number of previous companies |
| PercentSalaryHike | Numerical | Last salary hike percentage |
| PerformanceRating | Numerical (1–4) | Last performance review rating |
| RelationshipSatisfaction | Numerical (1–4) | Satisfaction with work relationships |
| StockOptionLevel | Numerical (0–3) | Stock option grant level |
| TotalWorkingYears | Numerical | Total years of professional experience |
| TrainingTimesLastYear | Numerical | Number of training sessions last year |
| WorkLifeBalance | Numerical (1–4) | Work-life balance rating |
| YearsAtCompany | Numerical | Years at current company |
| YearsInCurrentRole | Numerical | Years in current role |
| YearsSinceLastPromotion | Numerical | Years since last promotion |
| YearsWithCurrManager | Numerical | Years with current manager |
| Attrition | Categorical | TARGET — Yes/No |
| BusinessTravel | Categorical | Travel frequency |
| Department | Categorical | Employee's department |
| EducationField | Categorical | Field of education |
| Gender | Categorical | Male / Female |
| JobRole | Categorical | Job title / role |
| MaritalStatus | Categorical | Single / Married / Divorced |
| OverTime | Categorical | Whether employee works overtime (Yes/No) |

## 1.6 Initial Data Checks

The following standard exploratory checks were run on the raw dataset:

| Check | Code Used | Output / Finding |
|---|---|---|
| Dataset dimensions | df.shape | 1,470 rows × 35 columns |
| Data types & nulls | df.info() | All 35 columns fully populated — 0 null values |
| Statistical summary | df.describe() | Mean age ~37, mean monthly income ~6,500, varied distributions |
| Sample rows | df.head() | Confirmed mix of numerical and categorical columns |

## 1.7 Data Cleaning Summary

The following 4 columns were dropped as they provide zero predictive value:

| Column Removed | Reason for Removal |
|---|---|
| EmployeeNumber | Unique ID for each employee — purely an identifier, no pattern to learn |
| EmployeeCount | Constant value of 1 across all 1,470 rows — zero variance |
| Over18 | All employees are over 18 — single-value column, no information gain |
| StandardHours | Constant value of 80 for every row — zero variance |

After dropping these 4 columns, 31 features remained for analysis and modelling.

Duplicate rows: df.duplicated().sum() confirmed no duplicate records existed in the dataset.

## 1.8 Outlier Detection & Treatment

The IQR (Interquartile Range) method was applied to every numerical column to detect and remove extreme outliers. A custom function was written and looped over all numerical columns:

```
def outlier_detection(df, column):
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    return df[(df[column] >= lower_bound) & (df[column] <= upper_bound)]
```

| Parameter | Formula |
|---|---|
| Lower Bound | Q1 − (1.5 × IQR) |
| Upper Bound | Q3 + (1.5 × IQR) |
| Action taken | Rows with any numerical column outside bounds were removed |

This was applied iteratively across all numerical columns. The resulting cleaned dataset had fewer rows but no extreme outliers that could distort model training.

## 1.9 Exploratory Analysis Summary

**Univariate Analysis:**

- Attrition Count (countplot): Class imbalance present — majority of employees did NOT leave. Attrition='Yes' is the minority class, making it harder to predict.
- Age Distribution (histplot + KDE): Most employees are between 28–40 years old. Distribution is roughly bell-shaped with a slight right skew.
- Monthly Income Distribution (histplot + KDE): Right-skewed — most employees earn lower to moderate salaries, with a long tail of high earners.

**Bivariate Analysis:**

- Monthly Income vs Attrition (boxplot): Employees who left (Attrition=Yes) visibly earn lower monthly income compared to those who stayed.
- Overtime vs Attrition (countplot with hue): Employees working overtime show disproportionately higher attrition — a key finding.
- Job Satisfaction vs Attrition (countplot with hue): Employees with satisfaction rating 1 or 2 leave far more often than those rating 3 or 4.

**Key Observations:**

- Overtime is the single most visually striking driver of attrition in bivariate plots
- Lower income is consistently associated with higher probability of leaving
- Poor job satisfaction and work-life balance are strong secondary contributors

# 2. Feature Engineering Documentation

## 2.1 Categorical Encoding

After EDA, all categorical (object dtype) columns were encoded to numeric format using Label Encoding, so that scikit-learn models can process them. A LabelEncoder was fitted separately for each column and saved in a dictionary (le_d) for reuse on new data at prediction time.

| Column | Type | Encoded Values |
|---|---|---|
| Attrition | Binary | No → 0, Yes → 1 |
| OverTime | Binary | No → 0, Yes → 1 |
| Gender | Binary | Female → 0, Male → 1 |
| BusinessTravel | Ordinal | Non-Travel → 0, Travel_Frequently → 1, Travel_Rarely → 2 |
| Department | Nominal | Human Resources → 0, Research & Development → 1, Sales → 2 |
| EducationField | Nominal | Numeric label per field (Human Resources, Life Sciences, etc.) |
| JobRole | Nominal | Numeric label per role (Healthcare Rep, Manager, Sales Exec, etc.) |
| MaritalStatus | Nominal | Divorced → 0, Married → 1, Single → 2 |

Boolean values (True/False) that appeared after encoding were also mapped to 1/0 using df.replace({True: 1, False: 0}).

After encoding, df.dtypes confirmed all columns are int64 — fully numeric and ready for model input.

## 2.2 Feature Selection

Non-informative columns were removed prior to splitting:

- EmployeeNumber, EmployeeCount, Over18, StandardHours — all dropped (see Section 1.7)
- No multicollinearity check was applied at this stage — all remaining 30 features were retained in the model

## 2.3 Final Feature Set

| Component | Details |
|---|---|
| Input matrix X | All columns except Attrition — 30 features, all int64 |
| Target vector y | Attrition column — binary (0=No, 1=Yes) |
| Data shape | Cleaned, encoded, outlier-removed DataFrame fed into train_test_split |

# 3. Model Development and Evaluation Report

## 3.1 Dataset Splitting

The cleaned and encoded dataset was split using scikit-learn's train_test_split:

```
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)
```

| Split | Proportion | Purpose |
|---|---|---|
| Training set | 80% | Used to fit / train all three models |
| Testing set | 20% | Held-out set used only for final evaluation |
| random_state | 42 | Ensures reproducibility across runs |

## 3.2 Models Implemented

Three classification algorithms were selected — covering probabilistic, rule-based, and ensemble approaches:

| Model | Library | Why Selected |
|---|---|---|
| Naive Bayes (GaussianNB) | sklearn.naive_bayes.GaussianNB | Fast probabilistic baseline; handles numerical features with Gaussian assumption |
| Decision Tree | sklearn.tree.DecisionTreeClassifier | Interpretable rule-based model; captures non-linear patterns; easy to visualise |
| Random Forest | sklearn.ensemble.RandomForestClassifier | Ensemble of many Decision Trees; reduces overfitting via bagging; more robust |

## 3.3 Model Training

All three models were instantiated with default parameters and trained using the standard scikit-learn fit() API:

```
    nb = GaussianNB()
    rf = RandomForestClassifier()
    dt = DecisionTreeClassifier()
```

```
    nb.fit(X_train, y_train)
    rf.fit(X_train, y_train)
    dt.fit(X_train, y_train)
```

Predictions were then generated on the test set:

```
    nb_pred = nb.predict(X_test)
    rf_pred = rf.predict(X_test)
    dt_pred = dt.predict(X_test)
```

## 3.4 Model Evaluation Metrics

The following metrics from sklearn.metrics were used to evaluate each model on the test set:

| Metric | What it Measures |
|---|---|
| Accuracy Score | Overall % of correct predictions (both classes) |
| Confusion Matrix | Breakdown of TP, TN, FP, FN — shows what the model gets right and wrong |
| Misclassification Rate | 1 − Accuracy — proportion of incorrect predictions |

## 3.5 Results — Accuracy Scores

Actual output from the notebook (Cell 40):

```
Accuracy for Naive Bayes:   0.7285714285714285
Accuracy for Random Forest: 0.85
Accuracy for Decision Tree: 0.75
```

| Model | Accuracy | Misclassification Rate | Notes |
|---|---|---|---|
| Naive Bayes (GaussianNB) | 72.86% | 27.14% | Weakest — independence assumption likely violated in HR data |
| Decision Tree | 75.00% | 25.00% | Moderate — captures some patterns but may overfit on training data |
| Random Forest | 85.00% | 15.00% | Best — ensemble reduces variance and generalises well |

## 3.6 Confusion Matrix — Random Forest

Actual output from the notebook (Cell 37):

```
Confusion Matrix:
 [[115   2]
  [ 19   4]]
```

| | Predicted: No Attrition (0) | Predicted: Attrition (1) |
|---|---|---|
| Actual: No Attrition (0) | 115 — True Negatives (TN) | 2 — False Positives (FP) |
| Actual: Attrition (1) | 19 — False Negatives (FN) | 4 — True Positives (TP) |

Interpretation:

- TN = 115: Model correctly identified 115 employees who stayed — very high specificity
- TP = 4: Model correctly flagged 4 employees who actually left
- FP = 2: Only 2 employees were incorrectly flagged as leaving — very low false alarm rate

- FN = 19: 19 employees who left were missed — the model's main weakness due to class imbalance
- The high FN count reflects the class imbalance: 'Attrition=Yes' is a small minority, so the model is conservative about predicting it

## 3.7 Model Selection

**Selected Model: Random Forest Classifier**

| Reason | Detail |
|---|---|
| Highest accuracy | 85% vs 75% (Decision Tree) and 72.86% (Naive Bayes) |
| Better generalisation | Ensemble of many trees prevents the overfitting common in single Decision Trees |
| Handles feature mix | Robust to both the categorical (encoded) and numerical HR features |
| Noise tolerance | Averaging across trees makes it less sensitive to noisy or irrelevant features |
| Saved as best_model | best_model = rf — stored for use in live prediction pipeline |

## 3.8 Live Prediction — New Employee Example

The trained Random Forest model was tested on a new unseen employee profile (Cell 42–49):

| Feature | Value Entered |
|---|---|
| Age | 30 |
| MonthlyIncome | $4,000 |
| TotalWorkingYears | 5 years |
| YearsAtCompany | 3 years |
| DistanceFromHome | 10 km |
| JobSatisfaction | 2 (Low) |
| WorkLifeBalance | 2 (Poor) |
| OverTime | Yes |
| BusinessTravel | Travel_Rarely |
| Department | Sales |
| JobRole | Sales Executive |
| MaritalStatus | Single |
| Gender | Male |

Prediction output (Cell 47 & 49):

- Predicted class: 0 → Employee is likely to STAY (Attrition = No)
- Probability of Staying: 0.50 (50%)

- Probability of Leaving: 0.50 (50%)

Note: A 50/50 probability means this employee sits exactly on the decision boundary. Despite predicting 'Stay', this profile carries several high-risk signals (overtime, low satisfaction, low income, poor work-life balance) and should be flagged for a retention conversation.

# 4. Insights and Recommendations Report

## 4.1 Key Insights

The combined EDA and model analysis surfaced the following critical patterns in the HR dataset:

| Driver | Insight from Analysis |
|---|---|
| Overtime | The single strongest visual predictor — employees on overtime leave at dramatically higher rates |
| Monthly Income | Lower earners are far more likely to leave — visible in the Attrition vs MonthlyIncome boxplot |
| Job Satisfaction | Rating 1 or 2 out of 4 strongly correlates with Attrition=Yes across all departments |
| Work-Life Balance | Poor balance (1–2) increases attrition; employees value personal time as much as pay |
| Years at Company | Recently joined employees leave more — onboarding and early-career engagement are critical |
| Total Working Years | Less experienced employees churn more, suggesting early-career dissatisfaction or better outside offers |
| Distance from Home | Longer commutes add to daily stress and marginally increase attrition risk |
| Department (Sales) | Sales department shows relatively higher attrition — high-pressure environment with income variability |

## 4.2 Top Predictive Features

Based on the Random Forest model behaviour and EDA visualisations, the most important features driving predictions are:

| Rank | Feature | Contribution |
|---|---|---|
| 1 | OverTime | Highest impact — working overtime nearly doubles attrition risk |
| 2 | MonthlyIncome | Strong negative correlation — lower pay, higher risk |
| 3 | TotalWorkingYears | Fewer years experience = higher attrition probability |
| 4 | YearsAtCompany | Newer employees are at highest risk of leaving |
| 5 | JobSatisfaction | Scores 1–2 are a red flag for imminent attrition |
| 6 | Age | Younger employees switch companies more frequently |
| 7 | DistanceFromHome | Long commutes slowly erode retention |
| 8 | WorkLifeBalance | Scores 1–2 are associated with significantly higher attrition |

## 4.3 Business Recommendations

**1. Address Overtime Immediately**

- Overtime is the top predictor — cap overtime hours and monitor teams with high overtime rates

- Ensure employees working extra hours receive fair additional compensation or time-off in lieu
- Introduce flexible or hybrid work policies to reduce after-hours work pressure

**2. Competitive Compensation Review**

- Monthly income is the second strongest driver — underpaid employees are high flight risks
- Conduct a market benchmarking exercise, especially for Sales and junior-level roles
- Implement performance-linked salary reviews and transparent pay progression paths

**3. Improve Job Satisfaction & Engagement**

- Run regular pulse surveys — employees with satisfaction ratings of 1 or 2 need immediate intervention
- Invest in career development programs, internal mobility, and meaningful project assignments
- Managers should conduct regular 1-on-1s with low-satisfaction employees

**4. Strengthen Onboarding & Early-Career Retention**

- Employees with fewer years at the company leave most — first 1–3 years are the highest-risk window
- Build structured onboarding programs that create belonging and establish clear growth paths
- Pair new employees with mentors from their team or department

**5. Work-Life Balance Initiatives**

- Introduce wellness programs, mental health days, and remote work options
- Monitor department-level work-life balance scores and intervene when teams trend downward

## 4.4 Model Usage Guide

| Step | Action |
|---|---|
| Input | HR team provides a new employee's profile: age, income, overtime, satisfaction, etc. |
| Encoding | Apply saved le_d encoders to categorical columns before passing to model |
| Alignment | Align new data to X.columns to ensure all 30 features are present |
| Prediction | rf.predict(new_data) returns 0 (Stay) or 1 (Leave) |
| Probability | rf.predict_proba(new_data) gives confidence scores — flag employees > 60% risk |
| Action | HR team reviews flagged employees and initiates retention conversations |

## 4.5 Future Enhancements

- Apply SMOTE or class_weight='balanced' to address the class imbalance (minority Attrition=Yes class)
- Test advanced boosting models: XGBoost, LightGBM, CatBoost — likely to improve recall on the minority class
- Add external features: promotion history, engagement survey scores, peer review ratings
- Deploy the trained model as a REST API endpoint for integration with HR management platforms

- Build a real-time HR dashboard showing at-risk employees, department-level risk scores, and trend charts
- Schedule quarterly model retraining as new HR data becomes available

# 5. Conclusion

This capstone project successfully built an end-to-end machine learning pipeline to predict employee attrition using the HR-Employee-Attrition dataset (1,470 employees, 35 features).

The project followed a complete data science workflow — from raw data inspection and cleaning, through exploratory analysis, feature engineering and encoding, to model training, evaluation, and live prediction demonstration.

Three classification models were built and evaluated. The Random Forest classifier emerged as the clear winner with 85% accuracy on the held-out test set — outperforming Naive Bayes (72.86%) and Decision Tree (75%). The confusion matrix confirmed that the model has strong specificity (correctly identifying 115 out of 117 employees who stayed), though improving recall for the minority attrition class remains the primary area for future work.

The analysis confirmed that overtime, monthly income, job satisfaction, years at company, and work-life balance are the most powerful predictors of attrition. These findings translate directly into concrete, actionable HR strategies — from capping overtime and revising salaries, to improving engagement and designing better onboarding experiences.

The trained model is prediction-ready: given any new employee's profile, it returns an attrition prediction and a probability score — equipping HR teams with a data-driven early-warning system to retain talent before it is lost.

| Deliverable | Status | Detail |
|---|---|---|
| Data Cleaning | Done | Dropped 4 constant/ID columns; no nulls found in 1,470 records |
| Outlier Treatment | Done | IQR method applied iteratively across all numerical columns |
| Encoding | Done | Label encoding applied to 8 categorical columns; encoders saved in le_d |
| EDA | Done | Univariate (3 plots) + Bivariate (3 plots); key patterns identified |
| Model Training | Done | 3 models trained: Naive Bayes, Decision Tree, Random Forest |
| Model Evaluation | Done | Accuracy scores + confusion matrix for Random Forest |
| Best Model | Random Forest | Accuracy: 85% \| best_model = rf |
| Live Prediction | Done | New employee profile tested; output: Stay (50% / 50% borderline) |